Jonathan Zhu

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SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
Backend / Systems: FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth2, Pydantic, SQLAlchemy
Frontend: React, Next.js, TypeScript, Tailwind CSS, Material UI, React Query, Framer Motion
Databases: PostgreSQL, MySQL, MongoDB
Cloud / Infrastructure: AWS (EC2, RDS, S3, Bedrock, Cognito, API Gateway, Amplify, Lambda, Secrets Manager, VPC), WireGuard VPN
AI / ML: Amazon Nova Pro, Anthropic API, OpenAI API, Gemini API, Prompt Engineering, Multimodal LLMs
Tools & Compliance: Jenkins, Splunk, Dynatrace, HIPAA Compliance, BAA, pdf2image, asyncpg

WORK EXPERIENCE

Citywide Eye Care — Great Neck, NY
Full-Stack Software Engineer — *Jan 2024 – Present*

- Lead engineer for a production EMR platform used daily by clinical staff, architecting a zero-downtime migration from on-premise servers to AWS (EC2, RDS, S3, Cognito, API Gateway, Amplify)
- Owned HIPAA security architecture end to end: Cognito JWT auth, IP whitelisting, role-based access control, and WireGuard VPN, passing all external compliance reviews with zero audit findings
- Reduced physician documentation time by 75% by engineering structured charting workflows with auto-save, copy-forward, and S3-backed document storage via pre-signed URLs
- Partnered directly with clinicians to define requirements and iterate on features, accelerating feedback cycles and cutting rework across all major releases

ZLA (Client Contract – AI-Enabled CRM Platform) — Rochester, NY
Full-Stack Software Engineer — *Aug 2025 – Present*

- Built a production CRM from scratch on a 5-person Agile team, delivering lead management, property tracking, campaign workflows, and agent dashboards within tight contract timelines
- Engineered an AI-powered lead discovery pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI, cutting manual prospect research time by 60%
- Architected a normalized FastAPI and PostgreSQL backend with complex relational schemas; built a React/TypeScript frontend with Kanban pipelines and map-based lead search

Citywide Eye Care — Great Neck, NY
Software Engineer Intern — *Sep 2023 – Dec 2023*

- Built and deployed an AI health chatbot using Python and Gemini API, autonomously handling patient FAQs and symptom triage, cutting front-desk call volume by 50%
- Rebuilt the patient-facing website in Next.js and Material UI, improving SEO and driving a 25% increase in appointment conversion rates
- Implemented digital intake and scheduling workflows on AWS (EC2, S3, RDS, Secrets Manager), replacing all manual paper-based processes

ADP — Roseland, NJ
Application Developer Intern — *Jun 2023 – Aug 2023*

- Built an automated status-reporting tool adopted across the team, eliminating manual report generation and cutting decision turnaround time by 20%
- Reduced deployment time by 30% by optimizing the Jenkins CI/CD pipeline and improving MySQL query performance across internal analytics integrations
- Enhanced S3, Splunk, and Dynatrace integrations supporting internal analytics; recognized by engineering leadership for measurable impact on team velocity

PROJECTS

- **RemitAI:** HIPAA-compliant AI platform for healthcare RCM using Amazon Nova Pro on AWS Bedrock (BAA); processes EOB/ERA PDFs via multimodal pipeline, classifies denials by CARC/RARC codes, generates appeal strategies, and builds a self-learning payer knowledge base on async FastAPI, PostgreSQL, and React/TypeScript
- **AI Lead Discovery Pipeline:** Multi-source enrichment pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI to surface qualified real estate prospects, cutting manual research time by 60%
- **Enlighten Learning:** Full-stack nonprofit site built in React, TypeScript, and Material UI with serverless API integration (Google Apps Script), dark/light mode, animated UI, and donation tracking; supports tutoring programs for students in underserved communities across Long Island

EDUCATION

Georgia Institute of Technology — Admitted, Fall 2026
M.S. in Computer Science

Rochester Institute of Technology — Expected May 2026
B.S. in Software Engineering